

Math 49 Curriculum

Precalculus with Limits | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [1.2 Domain and Range](#); [1.5 Transformation of Functions](#); [4.1 Exponential Functions](#); [4.3 Logarithmic Functions](#); [4.6 Exponential and Logarithmic Equations](#)
Supplemental resources: [4.2 Graphs of Exponential Functions](#); [4.7 Exponential and Logarithmic Models](#); [4.4 Graphs of Logarithmic Functions](#)

UNIT LEARNING OBJECTIVES

- I can determine the domain and range of a function from a formula, graph, or table.
- I can describe and apply transformations of a parent function.
- I can analyze exponential functions using their equations, graphs, and contexts.
- I can analyze logarithmic functions using their equations, graphs, and contexts.
- I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED

Primary resources: [1.5 Transformation of Functions](#); [1.4 Composition of Functions](#); [1.7 Inverse Functions](#)

UNIT LEARNING OBJECTIVES

- I can determine whether a function is even, odd, or neither.
- I can add, subtract, multiply, and divide functions and state the resulting domain.
- I can evaluate a composition from formulas, graphs, or tables.
- I can write a formula for the composition of two functions and determine its domain.
- I can decompose a function into a composition of simpler functions.
- I can determine whether a function has an inverse on a given domain.
- I can find and verify an inverse function algebraically.
- I can interpret the relationship between a function and its inverse from graphs or tables.

Power Functions

REQUIRED

Primary resources: [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#); [College Algebra 2e 2.6 Other Types of Equations](#); [Precalculus 2e 3.8 Inverses and Radical Functions](#)
Supplemental resources: [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

- I can identify a power function and interpret the parameters in $y = kx^p$.
- I can construct a power function from two data points or a proportional relationship.
- I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
- I can recognize and model direct, inverse, and other power-variation relationships.
- I can solve equations involving integer or rational powers.
- I can build or select a power-function model for a context and interpret its parameters.
- I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Polynomial Functions

REQUIRED

Primary resources: [3.4 Graphs of Polynomial Functions](#); [3.6 Zeros of Polynomial Functions](#)
Supplemental resources: [3.6 Zeros of Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

- I can predict the end behavior of a polynomial from its degree and leading coefficient.
- I can determine polynomial zeros and their multiplicities.
- I can construct a polynomial function from specified zeros or graphical features.
- I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
- I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.
- I can build or select a polynomial model for a context and interpret its features.

Rational Functions and Limits

REQUIRED

Primary resources: [Precalculus 2e 3.7 Rational Functions](#); [Calculus Volume 1 2.2 The Limit of a Function](#); [Calculus Volume 1 2.3 The Limit Laws](#); [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)
Supplemental resources: [Calculus Volume 1 2.2 The Limit of a Function](#); [Precalculus 2e 3.4 Graphs of Polynomial Functions](#); [Precalculus 2e 3.7 Rational Functions](#); [Precalculus 2e 4.4 Graphs of Logarithmic Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#)

UNIT LEARNING OBJECTIVES

- I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
- I can sketch and analyze a rational function using its algebraic and asymptotic features.
- I can interpret limit notation as a statement about function behavior near an input.
- I can estimate a limit from a graph or table.
- I can evaluate a limit algebraically using direct substitution or simplification.
- I can interpret an infinite limit as vertical-asymptotic behavior.
- I can use limit notation to describe behavior at infinity.
- I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.
- I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Sequences and Series

REQUIRED

Primary resources: [11.1 Sequences and Their Notations](#); [11.2 Arithmetic Sequences](#); [11.3 Geometric Sequences](#); [11.4 Series and Their Notations](#)
Supplemental resources: [11.1 Sequences and Their Notations](#); [11.3 Geometric Sequences](#); [11.2 Arithmetic Sequences](#)

UNIT LEARNING OBJECTIVES

- I can evaluate and graph a sequence defined explicitly or recursively.
- I can write explicit and recursive formulas for arithmetic sequences.
- I can write explicit and recursive formulas for geometric sequences.
- I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
- I can create and interpret a sequence model for discrete change.
- I can interpret and use summation notation.
- I can calculate a finite arithmetic series.
- I can calculate a finite geometric series.
- I can determine whether an infinite geometric series converges and find its sum when it does.
- I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

Primary resources: [2.2 The Limit of a Function](#); [2.4 Continuity](#)

UNIT LEARNING OBJECTIVES

- I can determine one-sided limits and decide whether a two-sided limit exists.
- I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

Primary resources: [8.7 Parametric Equations: Graphs](#); [8.6 Parametric Equations](#)

UNIT LEARNING OBJECTIVES

- I can graph a plane curve defined by parametric equations with its orientation.
- I can eliminate a parameter to obtain a rectangular equation.
- I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

Primary resources: [10.1 The Ellipse](#); [10.2 The Hyperbola](#); [10.3 The Parabola](#)
Supplemental resources: [10.2 The Hyperbola](#); [10.3 The Parabola](#); [8.6 Parametric Equations](#); [10.1 The Ellipse](#)

UNIT LEARNING OBJECTIVES

- I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
- I can use completing the square to rewrite a general-form conic equation in standard form.
- I can write, interpret, and graph implicit and parametric equations of a circle.
- I can write, interpret, and graph implicit and parametric equations of an ellipse.
- I can write, interpret, and graph implicit and parametric equations of a hyperbola.

Conic Sections (Continued)

EXTENSION

UNIT LEARNING OBJECTIVES - CONTINUED

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

I can explain a parabola as the set of points equidistant from a focus and directrix.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** [1.2 Domain and Range](#); [1.5 Transformation of Functions](#); [4.1 Exponential Functions](#); [4.3 Logarithmic Functions](#); [4.6 Exponential and Logarithmic Equations](#)

Supplemental resources: [4.2 Graphs of Exponential Functions](#); [4.7 Exponential and Logarithmic Models](#); [4.4 Graphs of Logarithmic Functions](#)

M49-FND-001

I can determine the domain and range of a function from a formula, graph, or table.

OpenStax: **Primary resources:** [1.2 Domain and Range](#)

SUPPORTING SKILLS

D Determine domain restrictions from denominators, even roots, and logarithms.

SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31

R Read domain and range from a graph.

SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12

R Read domain and range from a table.

SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M49-FND-002

I can describe and apply transformations of a parent function.

OpenStax: **Primary resources:** [1.5 Transformation of Functions](#)

SUPPORTING SKILLS

R Interpret transformations applied to function inputs.

SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31

R Distinguish stretches from compressions and horizontal effects from vertical effects.

SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31

R Interpret transformations applied to function outputs.

SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

I can analyze exponential functions using their equations, graphs, and contexts.

OpenStax: **Primary resources:** [4.1 Exponential Functions](#)

Supplemental resources: [4.2 Graphs of Exponential Functions](#); [4.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- R** Identify horizontal asymptotic behavior of an exponential graph.

SK-EXP-ASYMPTOTE · first introduced in Math 31

- R** Identify initial amount, rate, and time interval in context.

SK-EXP-MODEL-CONTEXT · first introduced in Math 31

- A** Connect parameters in a formula to features of its graph.

SK-REP-FORMULA-GRAPH · first introduced in Math 21

I can analyze logarithmic functions using their equations, graphs, and contexts.

OpenStax: **Primary resources:** [4.3 Logarithmic Functions](#)

Supplemental resources: [4.4 Graphs of Logarithmic Functions](#); [4.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- R** Identify vertical asymptotic behavior of a logarithmic graph.

SK-LOG-ASYMPTOTE · first introduced in Math 31

- R** Interpret a logarithm as an exponent.

SK-LOG-DEFINITION · first introduced in Math 31

- A** Connect parameters in a formula to features of its graph.

SK-REP-FORMULA-GRAPH · first introduced in Math 21

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

OpenStax: **Primary resources:** [4.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- R** Rewrite exponential expressions using a common base when possible.

SK-EXP-REWRITE-BASE · first introduced in Math 31

- A** Reject solutions that make a logarithm argument nonpositive.

SK-LOG-DOMAIN-CHECK · first introduced in Math 31

- A** Apply the logarithm power property in either direction.

SK-LOG-POWER · first introduced in Math 31

- A** Apply the logarithm product property in either direction.

SK-LOG-PRODUCT · first introduced in Math 31

- A** Apply the logarithm quotient property in either direction.

SK-LOG-QUOTIENT · first introduced in Math 31

- R** Isolate an exponential expression and apply a logarithm.

SK-LOG-SOLVE-EXP · first introduced in Math 31

- R** Combine or isolate logarithms before converting to exponential form.

SK-LOG-SOLVE-LOG · first introduced in Math 31

Function Operations, Composition, and Inverses

REQUIRED

OpenStax coverage: **Primary resources:** [1.5 Transformation of Functions](#); [1.4 Composition of Functions](#); [1.7 Inverse Functions](#)

M49-FND-003

I can determine whether a function is even, odd, or neither.

OpenStax: **Primary resources:** [1.5 Transformation of Functions](#)

SUPPORTING SKILLS

- D** Test a formula or graph for even or odd symmetry.
SK-FUNC-SYMMETRY · first introduced in Math 31

M49-OPS-001

I can add, subtract, multiply, and divide functions and state the resulting domain.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Determine the domain of a sum, difference, product, quotient, or composition.
SK-FUNC-DOMAIN-COMBINE
- I** Add, subtract, multiply, and divide function outputs.
SK-FUNC-OPERATIONS

M49-OPS-002

I can evaluate a composition from formulas, graphs, or tables.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Follow intermediate values to compose functions from tables.
SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- R** Read and write notation for function composition and inverse functions.
SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M49-OPS-003

I can write a formula for the composition of two functions and determine its domain.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Determine the domain of a sum, difference, product, quotient, or composition.
SK-FUNC-DOMAIN-COMBINE · first introduced in Math 39

M49-OPS-004

I can decompose a function into a composition of simpler functions.

OpenStax: **Primary resources:** [1.4 Composition of Functions](#)

SUPPORTING SKILLS

- A** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I** Identify simpler inner and outer functions in a composition.
SK-FUNC-DECOMPOSE

M49-OPS-005

I can determine whether a function has an inverse on a given domain.

OpenStax: **Primary resources:** [1.7 Inverse Functions](#)

SUPPORTING SKILLS

- D** Apply the horizontal line test for invertibility.
SK-FUNC-INVERSE-HLT · first introduced in Math 31
- R** Test whether a function is one-to-one on a domain.
SK-FUNC-ONE-TO-ONE · first introduced in Math 31

M49-OPS-006

I can find and verify an inverse function algebraically.

OpenStax: **Primary resources:** [1.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Exchange variables and solve to obtain an inverse formula.
SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D** Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY · first introduced in Math 31

M49-OPS-007

I can interpret the relationship between a function and its inverse from graphs or tables.

OpenStax: **Primary resources:** [1.7 Inverse Functions](#)

SUPPORTING SKILLS

- D** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D** Reverse input and output units when interpreting an inverse.
SK-FUNC-INVERSE-UNITS · first introduced in Math 31

Power Functions

REQUIRED

OpenStax coverage: **Primary resources:** [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#); [College Algebra 2e 2.6 Other Types of Equations](#); [Precalculus 2e 3.8 Inverses and Radical Functions](#)
Supplemental resources: [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)

M49-PWR-001

I can identify a power function and interpret the parameters in $y = kx^p$.

OpenStax: **Primary resources:** [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- D** Interpret the coefficient and exponent in a power function.
SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
 - R** Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
SK-POWER-VOCABULARY · first introduced in Math 39
 - D** Represent and interpret a power-variation relationship.
SK-VAR-POWER · first introduced in Math 39
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M49-PWR-002

I can construct a power function from two data points or a proportional relationship.

OpenStax: **Primary resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
 - D** Determine a power function from two points or a proportional relationship.
SK-POWER-CONSTRUCT · first introduced in Math 39
-

M49-PWR-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

OpenStax: **Primary resources:** [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- D** Determine power-function behavior as the input approaches zero or positive or negative infinity.
SK-POWER-END-BEHAVIOR · first introduced in Math 39
 - D** Relate the coefficient and exponent of a power function to its graph shape.
SK-POWER-SHAPE · first introduced in Math 39
 - A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21
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M49-PWR-004

I can recognize and model direct, inverse, and other power-variation relationships.

OpenStax: **Primary resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- D** Represent and interpret a direct-variation relationship.
SK-VAR-DIRECT · first introduced in Math 39
- D** Represent and interpret an inverse-variation relationship.
SK-VAR-INVERSE · first introduced in Math 39
- D** Represent and interpret a power-variation relationship.
SK-VAR-POWER · first introduced in Math 39

I can solve equations involving integer or rational powers.

OpenStax: **Primary resources:** [2.6 Other Types of Equations](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21

I can build or select a power-function model for a context and interpret its parameters.

OpenStax: **Primary resources:** [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- I** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Determine a power function from two points or a proportional relationship.
SK-POWER-CONSTRUCT · first introduced in Math 39

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

OpenStax: **Primary resources:** [3.8 Inverses and Radical Functions](#)

Supplemental resources: [3.3 Power Functions and Polynomial Functions](#)

SUPPORTING SKILLS

- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21
- A** Require an even-root radicand to be nonnegative when determining domain.
SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- I** Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent.
SK-POWER-FRACTIONAL-ANALYZE

Polynomial Functions

REQUIRED

OpenStax coverage: **Primary resources:** [3.4 Graphs of Polynomial Functions](#); [3.6 Zeros of Polynomial Functions](#)
Supplemental resources: [3.6 Zeros of Polynomial Functions](#)

M49-ALG-002

I can predict the end behavior of a polynomial from its degree and leading coefficient.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- I** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR
- R** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M49-ALG-003

I can determine polynomial zeros and their multiplicities.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- I** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12
- D** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-004

I can construct a polynomial function from specified zeros or graphical features.

OpenStax: **Primary resources:** [3.6 Zeros of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT · first introduced in Math 31
- A** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-005

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- I** Use inflection points to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-INFLECTION
- I** Use turning points to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-TURNING
- I** Use zeros and their multiplicities to establish a minimum possible polynomial degree.
SK-POLY-DEGREE-ZEROS
- A** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39

I can build or select a polynomial model for a context and interpret its features.

OpenStax: **Primary resources:** [3.4 Graphs of Polynomial Functions](#)

Supplemental resources: [3.6 Zeros of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT · first introduced in Math 31

Rational Functions and Limits

REQUIRED

OpenStax coverage: **Primary resources:** [Precalculus 2e 3.7 Rational Functions](#); [Calculus Volume 1 2.2 The Limit of a Function](#); [Calculus Volume 1 2.3 The Limit Laws](#); [Precalculus 2e 3.3 Power Functions and Polynomial Functions](#)
Supplemental resources: [Calculus Volume 1 2.2 The Limit of a Function](#); [Precalculus 2e 3.4 Graphs of Polynomial Functions](#); [Precalculus 2e 3.7 Rational Functions](#); [Precalculus 2e 4.4 Graphs of Logarithmic Functions](#); [Precalculus 2e 3.9 Modeling Using Variation](#)

M49-ALG-006

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

OpenStax: **Primary resources:** [3.7 Rational Functions](#)

SUPPORTING SKILLS

- A** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- I** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR
- A** Factor numerators and denominators completely.
SK-RAT-FACTOR · first introduced in Math 21
- I** Recall and correctly use rational-function terms including restriction, hole, asymptote, and end behavior.
SK-RAT-FUNCTION-VOCAB
- I** Identify a removable discontinuity and its coordinates.
SK-RAT-HOLE
- I** Determine vertical asymptotes from noncanceling denominator zeros.
SK-RAT-VERTICAL-ASYMPTOTE
- R** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY · first introduced in Math 21

M49-ALG-007

I can sketch and analyze a rational function using its algebraic and asymptotic features.

OpenStax: **Primary resources:** [3.7 Rational Functions](#)

SUPPORTING SKILLS

- D** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR · first introduced in Math 49
- D** Identify a removable discontinuity and its coordinates.
SK-RAT-HOLE · first introduced in Math 49
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21
- I** Determine the sign of a rational function on intervals.
SK-RAT-SIGN-INTERVAL
- D** Determine vertical asymptotes from noncanceling denominator zeros.
SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-001

I can interpret limit notation as a statement about function behavior near an input.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Read and write limit notation with correct approach direction.
SK-LIM-NOTATION

M49-LIM-002

I can estimate a limit from a graph or table.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Estimate one-sided and two-sided limits from a graph.
SK-LIM-GRAPH
- I** Estimate a limit from input-output values approaching from both sides.
SK-LIM-TABLE

M49-LIM-003

I can evaluate a limit algebraically using direct substitution or simplification.

OpenStax: **Primary resources:** [2.3 The Limit Laws](#)

SUPPORTING SKILLS

- I** Resolve an indeterminate algebraic form by factoring or rationalizing.
SK-LIM-SIMPLIFY
- I** Test a limit using direct substitution.
SK-LIM-SUBSTITUTE
- A** Factor numerators and denominators completely.
SK-RAT-FACTOR · first introduced in Math 21

M49-LIM-005

I can interpret an infinite limit as vertical-asymptotic behavior.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Interpret unbounded output near a finite input as an infinite limit.
SK-LIM-INFINITE
- D** Determine vertical asymptotes from noncanceling denominator zeros.
SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-006

I can use limit notation to describe behavior at infinity.

OpenStax: **Primary resources:** [Precalculus 2e 3.7 Rational Functions](#)
Supplemental resources: [Calculus Volume 1 2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- I** Compare dominant terms to determine behavior at infinity.
SK-LIM-INFINITY
- D** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR · first introduced in Math 49

I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.

OpenStax: **Primary resources:** [3.3 Power Functions and Polynomial Functions](#)

Supplemental resources: [3.4 Graphs of Polynomial Functions](#); [3.7 Rational Functions](#); [4.4 Graphs of Logarithmic Functions](#)

SUPPORTING SKILLS

- A** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- D** Compare the long-run growth or decay of functions from different families.
SK-FUNC-DOMINANCE · first introduced in Math 39
- I** Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions.
SK-LIM-GROWTH-ORDER
- A** Compare dominant terms to determine behavior at infinity.
SK-LIM-INFINITY · first introduced in Math 49

I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

OpenStax: **Primary resources:** [3.7 Rational Functions](#)

Supplemental resources: [3.9 Modeling Using Variation](#)

SUPPORTING SKILLS

- D** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Determine horizontal or slant asymptotic behavior.
SK-RAT-END-BEHAVIOR · first introduced in Math 49

Sequences and Series

REQUIRED

OpenStax coverage: **Primary resources:** [11.1 Sequences and Their Notations](#); [11.2 Arithmetic Sequences](#); [11.3 Geometric Sequences](#); [11.4 Series and Their Notations](#)

Supplemental resources: [11.1 Sequences and Their Notations](#); [11.3 Geometric Sequences](#); [11.2 Arithmetic Sequences](#)

M49-SEQ-001

I can evaluate and graph a sequence defined explicitly or recursively.

OpenStax: **Primary resources:** [11.1 Sequences and Their Notations](#)

SUPPORTING SKILLS

- I** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT
- I** Graph a sequence as discrete ordered pairs.
SK-SEQ-GRAPH
- I** Interpret subscripts and evaluate terms of a sequence.
SK-SEQ-NOTATION
- I** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE
- I** Recall and correctly use the terms sequence, term, index, explicit formula, recursive formula, arithmetic sequence, and geometric sequence.
SK-SEQ-VOCABULARY

M49-SEQ-002

I can write explicit and recursive formulas for arithmetic sequences.

OpenStax: **Primary resources:** [11.2 Arithmetic Sequences](#)

SUPPORTING SKILLS

- I** Recognize constant difference in an arithmetic sequence.
SK-SEQ-DIFFERENCE
- D** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT · first introduced in Math 49
- D** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-003

I can write explicit and recursive formulas for geometric sequences.

OpenStax: **Primary resources:** [11.3 Geometric Sequences](#)

SUPPORTING SKILLS

- D** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT · first introduced in Math 49
- I** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO
- D** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-004

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

OpenStax: **Primary resources:** [11.2 Arithmetic Sequences](#)

Supplemental resources: [11.1 Sequences and Their Notations](#); [11.3 Geometric Sequences](#)

SUPPORTING SKILLS

- A** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- D** Recognize constant difference in an arithmetic sequence.
SK-SEQ-DIFFERENCE · first introduced in Math 49
- D** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO · first introduced in Math 49

M49-SEQ-005

I can create and interpret a sequence model for discrete change.

OpenStax: **Primary resources:** [11.3 Geometric Sequences](#)

Supplemental resources: [11.2 Arithmetic Sequences](#)

SUPPORTING SKILLS

- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Generate an explicit formula from a sequence pattern.
SK-SEQ-EXPLICIT · first introduced in Math 49
- I** Identify initial value and pattern of change in a discrete context.
SK-SEQ-MODEL
- A** Generate a recursive formula with an initial condition.
SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SER-001

I can interpret and use summation notation.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Interpret subscripts and evaluate terms of a sequence.
SK-SEQ-NOTATION · first introduced in Math 49
- I** Expand and evaluate a finite sum written in sigma notation.
SK-SER-SIGMA
- I** Recall and correctly use the terms series, partial sum, sigma notation, finite series, infinite series, and convergence.
SK-SER-VOCABULARY

M49-SER-002

I can calculate a finite arithmetic series.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Recognize constant difference in an arithmetic sequence.
SK-SEQ-DIFFERENCE · first introduced in Math 49
- I** Apply an arithmetic-series sum formula.
SK-SER-ARITHMETIC

M49-SER-003

I can calculate a finite geometric series.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO · first introduced in Math 49
- I** Apply a finite geometric-series sum formula.
SK-SER-GEOMETRIC-FINITE

M49-SER-004

I can determine whether an infinite geometric series converges and find its sum when it does.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Recognize constant ratio in a geometric sequence.
SK-SEQ-RATIO · first introduced in Math 49
- I** Test convergence and sum an infinite geometric series.
SK-SER-GEOMETRIC-INFINITE

M49-SER-005

I can create and interpret a series model for accumulated change.

OpenStax: **Primary resources:** [11.4 Series and Their Notations](#)

SUPPORTING SKILLS

- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Apply an arithmetic-series sum formula.
SK-SER-ARITHMETIC · first introduced in Math 49
- A** Apply a finite geometric-series sum formula.
SK-SER-GEOMETRIC-FINITE · first introduced in Math 49
- A** Test convergence and sum an infinite geometric series.
SK-SER-GEOMETRIC-INFINITE · first introduced in Math 49
- I** Identify the initial term, change factor, and number of terms in an application.
SK-SER-MODEL

Additional Limit Concepts

EXTENSION

OpenStax coverage: **Primary resources:** [2.2 The Limit of a Function](#); [2.4 Continuity](#)

M49-LIM-004

I can determine one-sided limits and decide whether a two-sided limit exists.

OpenStax: **Primary resources:** [2.2 The Limit of a Function](#)

SUPPORTING SKILLS

- D** Estimate one-sided and two-sided limits from a graph.
SK-LIM-GRAPH · first introduced in Math 49
- D** Read and write limit notation with correct approach direction.
SK-LIM-NOTATION · first introduced in Math 49
- A** Estimate a limit from input-output values approaching from both sides.
SK-LIM-TABLE · first introduced in Math 49

M49-LIM-008

I can distinguish a function value from a limit and determine continuity at a point.

OpenStax: **Primary resources:** [2.4 Continuity](#)

SUPPORTING SKILLS

- I** Check the three conditions for continuity at a point.
SK-LIM-CONTINUITY
- A** Estimate one-sided and two-sided limits from a graph.
SK-LIM-GRAPH · first introduced in Math 49
- D** Test a limit using direct substitution.
SK-LIM-SUBSTITUTE · first introduced in Math 49

Parametric Equations

EXTENSION

OpenStax coverage: **Primary resources:** [8.7 Parametric Equations: Graphs](#); [8.6 Parametric Equations](#)

M49-PAR-001

I can graph a plane curve defined by parametric equations with its orientation.

OpenStax: **Primary resources:** [8.7 Parametric Equations: Graphs](#)

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- D** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

M49-PAR-002

I can eliminate a parameter to obtain a rectangular equation.

OpenStax: **Primary resources:** [8.6 Parametric Equations](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Eliminate a parameter algebraically.
SK-PAR-ELIMINATE · first introduced in Math 32
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M49-PAR-003

I can create or interpret parametric equations for motion in the plane.

OpenStax: **Primary resources:** [8.7 Parametric Equations: Graphs](#)

SUPPORTING SKILLS

- D** Interpret or construct parametric equations for planar motion.
SK-PAR-MODEL · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Conic Sections

EXTENSION

OpenStax coverage: **Primary resources:** [10.1 The Ellipse](#); [10.2 The Hyperbola](#); [10.3 The Parabola](#)

Supplemental resources: [10.2 The Hyperbola](#); [10.3 The Parabola](#); [8.6 Parametric Equations](#); [10.1 The Ellipse](#)

M49-CON-001

I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [10.2 The Hyperbola](#)

SUPPORTING SKILLS

- I** Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
SK-CONIC-CLASSIFY
 - I** Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.
SK-CONIC-DISTANCE-DEFINITION
 - I** Recall the defining terms and standard notation for circles, parabolas, ellipses, and hyperbolas.
SK-CONIC-VOCABULARY
-

M49-CON-002

I can use completing the square to rewrite a general-form conic equation in standard form.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [10.2 The Hyperbola](#); [10.3 The Parabola](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
 - I** Complete squares to convert a general conic equation to standard form.
SK-CONIC-GENERAL-STANDARD
 - D** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
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M49-CON-003

I can write, interpret, and graph implicit and parametric equations of a circle.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [8.6 Parametric Equations](#)

SUPPORTING SKILLS

- I** Connect implicit, standard, and parametric equations of a circle.
SK-CONIC-CIRCLE-EQUATIONS
- I** Determine and use a circle's center and radius.
SK-CONIC-CIRCLE-FEATURES
- A** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

I can write, interpret, and graph implicit and parametric equations of an ellipse.OpenStax: **Primary resources:** [10.1 The Ellipse](#)**Supplemental resources:** [8.6 Parametric Equations](#)**SUPPORTING SKILLS**

- I** Connect implicit, standard, and parametric equations of an ellipse.
SK-CONIC-ELLIPSE-EQUATIONS
- I** Determine and use an ellipse's center, axes, vertices, and foci.
SK-CONIC-ELLIPSE-FEATURES
- A** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

I can write, interpret, and graph implicit and parametric equations of a hyperbola.OpenStax: **Primary resources:** [10.2 The Hyperbola](#)**Supplemental resources:** [8.6 Parametric Equations](#)**SUPPORTING SKILLS**

- I** Connect implicit, standard, and parametric equations of a hyperbola.
SK-CONIC-HYPERBOLA-EQUATIONS
- I** Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
SK-CONIC-HYPERBOLA-FEATURES
- A** Graph a parametric curve with its direction of motion.
SK-PAR-GRAPH · first introduced in Math 32
- A** Generate ordered pairs from parametric equations.
SK-PAR-TABLE · first introduced in Math 32

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.OpenStax: **Primary resources:** [10.2 The Hyperbola](#)**Supplemental resources:** [10.1 The Ellipse](#)**SUPPORTING SKILLS**

- D** Determine and use a circle's center and radius.
SK-CONIC-CIRCLE-FEATURES · first introduced in Math 49
- A** Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
SK-CONIC-CLASSIFY · first introduced in Math 49
- D** Determine and use an ellipse's center, axes, vertices, and foci.
SK-CONIC-ELLIPSE-FEATURES · first introduced in Math 49
- D** Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
SK-CONIC-HYPERBOLA-FEATURES · first introduced in Math 49

M49-CON-007

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

OpenStax: **Primary resources:** [10.1 The Ellipse](#)

Supplemental resources: [10.2 The Hyperbola](#)

SUPPORTING SKILLS

- D** Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.

SK-CONIC-DISTANCE-DEFINITION · first introduced in Math 49

- A** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.

SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22

M49-CON-008

I can explain a parabola as the set of points equidistant from a focus and directrix.

OpenStax: **Primary resources:** [10.3 The Parabola](#)

SUPPORTING SKILLS

- I** Explain a parabola using its focus-directrix distance relationship.

SK-CONIC-PARABOLA-DEFINITION

- A** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.

SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22